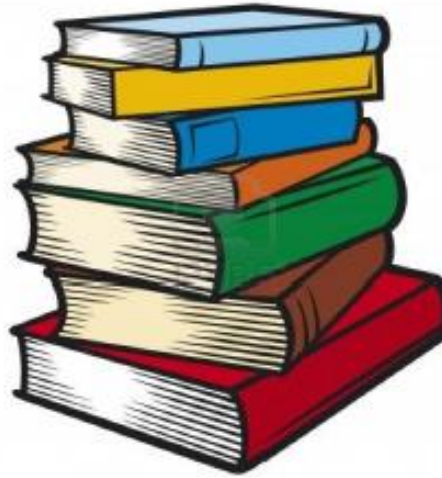


EE204 Introduction to Semiconductor Devices



Logistics



Books



**Syllabus &
Grading**

Logistics

Class Location and Time:

□ Tuesday, 14:00-15:50 , Rm 304, 1st teaching building

Instructors:

A/P Aung Ko Ko Kyaw (Alex)

aung@sustech.edu.cn

Rm 547, Engineering Building, South Tower

Tel. (o): 8801 8531

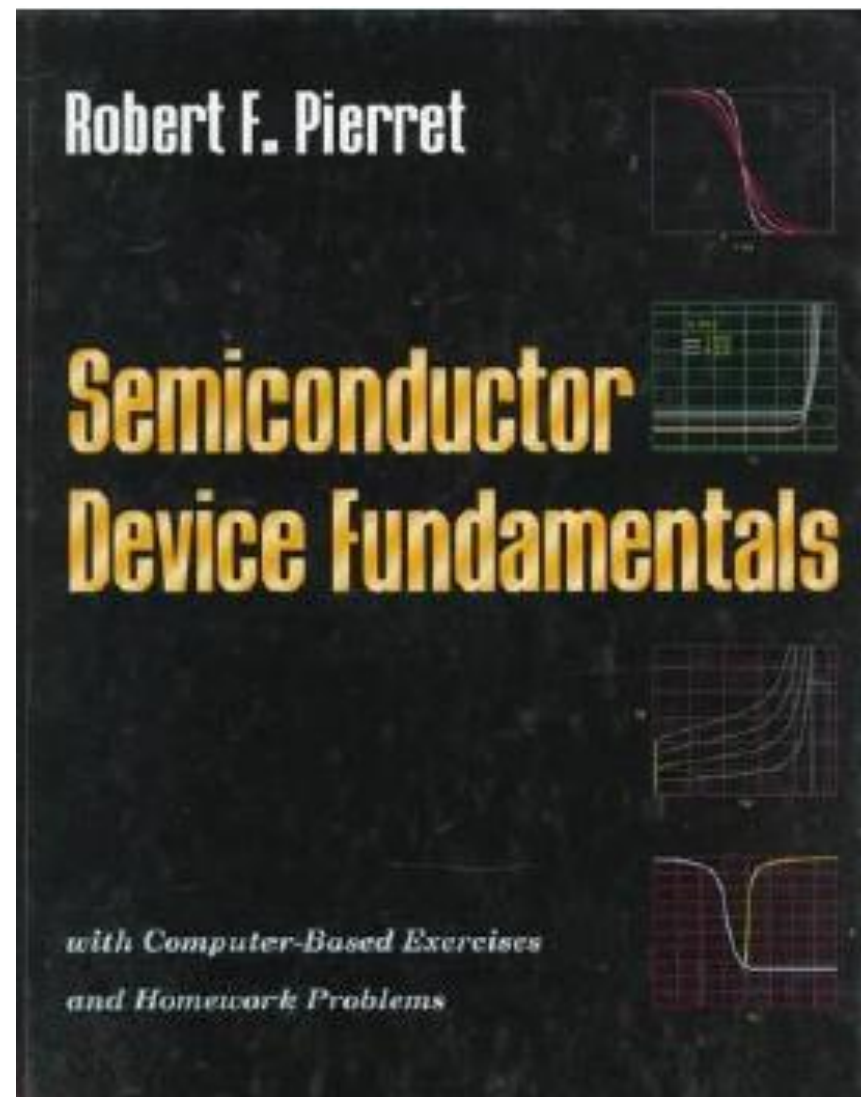
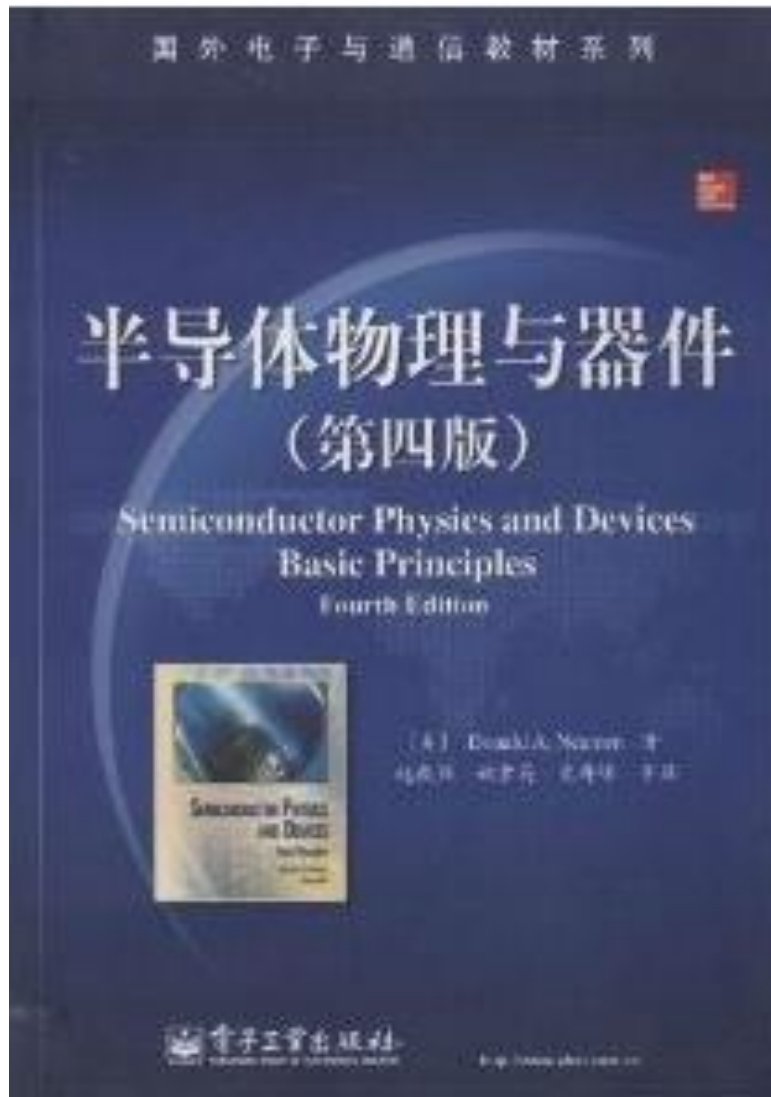
TA:

徐鑫 (12332199@mail.sustech.edu.cn)

Lab:

徐琳琳

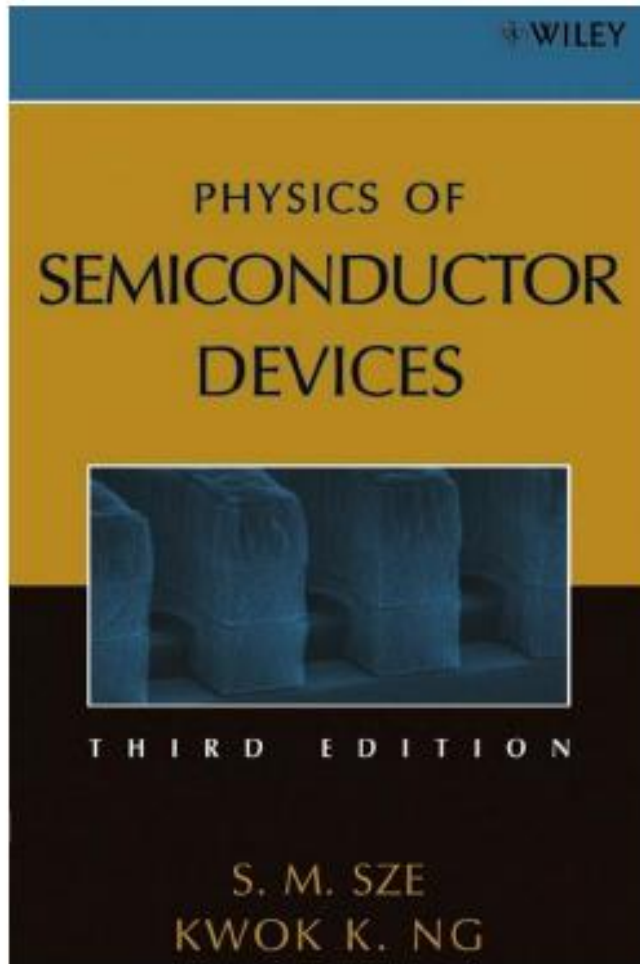
Book



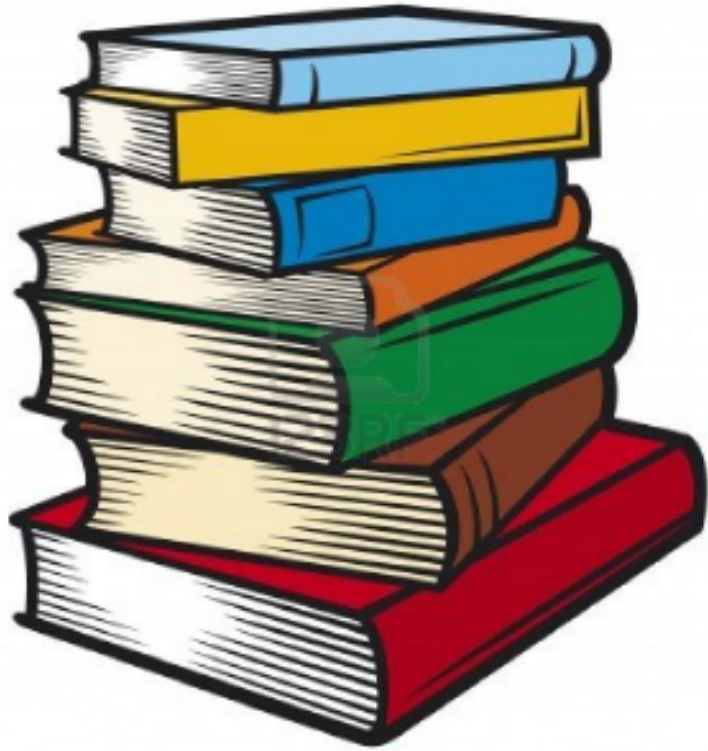


Books : Advanced / Reference

Bible of all devices



Books



- Browse through the 3 and follow any one you like
- Some of the problems in the problem sets will be taken from the books but you do not need to have the book to solve them
- Follow **at least one** book along with the class note

Tentative schedule

1st -8th week

Chapter 1-4. Crystal structure, Energy band, and Semiconductor in equilibrium (3-class hour)

Chapter 5. Carrier transport (1-class hour)

Chapter 6. Non-equilibrium excess carrier (1-class hour)

Chapter 7. PN junction (1-class hour)

Chapter 8. PN diode (2-class hour)

Midterm

9th-16th week

Chapter 9. Metal-Semiconductor contact (2-class hour)

Chapter 10. Bipolar Junction Transistors (3-class hour)

Chapter 11. Metal-oxide-semiconductor (MOS) capacitor (2-class hour)

Chapter 12. MOSFET (1-class hour)

Final exam

Lab and project

Simulation (Silvaco) and testing the I-V characteristics of

- 1). p-n diodes
- 2). solar cells
- 3). MOSFETs.

Grading

Lab 20%, Midterm 20%, Final 40%, homework (10%) and class participation (10%)

Student responsibilities and policies

- ✓ Attending classes;
- ✓ Reading textbook;
- ✓ Completing homework assignments in time; Late homework will not be accepted;
- ✓ Collaboration policy: Limited homework collaboration. You are welcome to discuss problems and concepts with fellow students verbally, but you should not share written work;
- ✓ Exams will be “open note” (1 page)